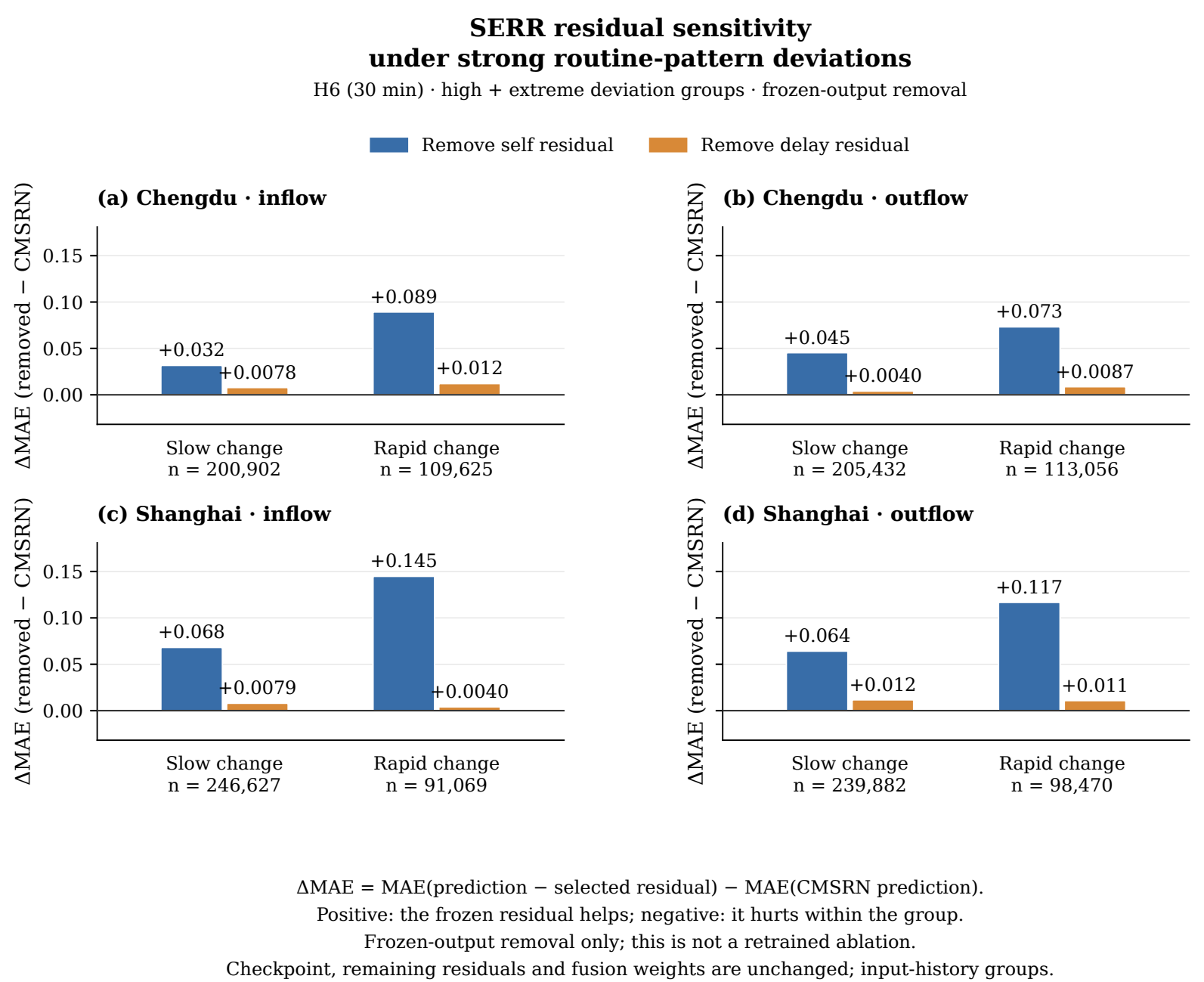

SERR residual sensitivity
under strong routine-pattern deviations
H6 (30 min) · high + extreme deviation groups · frozen-output removal
Remove self residual
Remove delay residual
(a) Chengdu · inflow
(b) Chengdu · outflow
(c) Shanghai · inflow
(d) Shanghai · outflow
ΔMAE (removed − CMSRN)
0.15
0.10
0.05
0.00
+0.032
+0.0078
+0.089
+0.012
+0.045
+0.0040
+0.073
+0.0087
+0.068
+0.0079
+0.145
+0.0040
+0.064
+0.012
+0.117
+0.011
Slow change
n = 200,902
Rapid change
n = 109,625
Slow change
n = 205,432
Rapid change
n = 113,056
Slow change
n = 246,627
Rapid change
n = 91,069
Slow change
n = 239,882
Rapid change
n = 98,470
ΔMAE = MAE(prediction − selected residual) − MAE(CMSRN prediction).
Positive: the frozen residual helps; negative: it hurts within the group.
Frozen-output removal only; this is not a retrained ablation.
Checkpoint, remaining residuals and fusion weights are unchanged; input-history groups.